NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.5a Non-convective extreme-wind design speeds: AI climate emulators

Group AI Weather & Climate Models for Wind Engineering

Status Shelved

Period 2026-10-14 14:21 — 2026-10-18 16:54

Stage conclusion

Project report

Non-convective extreme wind design wind speeds: AI climate emulators

Objective

Use AI climate emulators to generate hundreds to thousands of synthetic years, extract station (or grid-point) annual maximum winds, and estimate long-return-period design wind speeds for non-thunderstorm (extratropical cyclone, cold front) extreme winds, filling the synoptic-scale component of the mixed-climate framework in the EVA track. Step 1 only answers: whether the emulator's annual maximum wind distribution is systematically biased low, and whether statistical downscaling can rescue it. G (design wind speeds under climate change) is Stage 2; once C passes, start with "cold-era training, warm-era validation".

Current Status

10-17 Shelved: Station-level track Shelved; grid-cell-level conclusions retained (set by root conversation, upheld after revisions per review verdict; proposal). - Reason for shelving: Station v1 (same-season pairing) failed and v2 (distributional leave-one-station-out transfer across 29 new stations) was inconclusive; under both protocols and the spike-removed subset, ERA5 grid-cell information showed **no gain over the constant baseline** for station non-convective cold-season extremes (SS ≈ 0, hit rate tied with baseline). This does not "prove non-transferability"; the v2 confidence interval is also compatible with "moderate skill that went undetected." The v2 primary dataset still contains observational contamination including 100–291 kn spikes at 8 stations, not yet cleaned. - Retained grid-cell conclusions (narrowed per Verdict Clause 6): Unseen grid-cell GEV50 narrowly passed (2/7 outside the 10-member range, threshold ≤2), while POT50 and p90 in the same round each had 3/7 outside the range, and the initial 8-cell primary criterion was "inconclusive." Plains cells are inconsistent: KFSD −23%, KDSM −29%, and KOMA −23% are biased low, whereas KFAR +4.6% and KSTL +9.4% are biased high (KDSM and KOMA figures come from the previously inspected 10-member round). We can only state that "some Plains grid cells are biased low, with inconsistent signs across the region." - Restart conditions: (a) Pre-registered observational QC re-check—already performed once on 10-17: the frozen protocol mechanically "passed," but once PK WND is exempted from Q3 the SS confidence interval crosses zero, meaning the pass depends on the Q3 specification; therefore it remains Shelved with no further re-tests, and any new restart requires new stations or longer records; (b) synoptic-event-level pairing under a new pre-registration; (c) station exposure correction (directional roughness, obstacles)—constitutes changing the scientific question, requires Liang Hu's decision, and is not pursued here; we do not assert that exposure is the primary cause. - G remains deferred.

Previous Status Log

Started 10-14. Three tasks running in parallel: (1) ideas is verifying AI climate emulators runnable on Colab A100 (ACE2, NeuralGCM, ArchesWeatherGen, DLESyM, cBottle, etc.), selecting one to run ≥30 simulated years and comparing annual maximum winds against ERA5 at 6–10 grid points in the central/northeastern US; (2) asking the EVA Lead whether ready-made non-thunderstorm ASOS annual maximum winds exist; (3) Grok quick check returned and verified on 10-14 14:25: ACE2-ERA5 natively outputs 10 m wind at 1°/6 h under Apache-2.0 and is the top candidate; non-tropical-cyclone wind extremes have not been evaluated by anyone; "ML models underestimate extratropical cyclone peak winds" is pending verification. Two literature reports have been verified and sent for review, and the review verdict has been issued (all 3 reviewers support with reservations; C continues). The literature gap is narrowed to: no prior work found using an ERA5-trained AI climate emulator to estimate 100–1,000-year return levels of near-surface 10 m non-convective, non-tropical-cyclone winds linked to station 3 s gust design wind speeds; adjacent precedents include Singh 2026 (850 hPa wind return levels, generative outperforms deterministic), Kwa 2026 (HiRO-ACEv2 10 m wind tails in a model world), LUCIE (long rollouts overestimate extreme frequency for most variables), and Paciorek 2026 (ACE2 10,000-year ensemble, precipitation/temperature only). Protocol revisions have been sent to ideas.

Validation Protocol (Pre-specified 10-14, Revised After Review)

The event population is unified as "non-convective, non-tropical-cyclone": at the grid level, take cold-season (October–March) maximum winds at grid points unaffected by tropical cyclones; at the station level, exclude thunderstorms and tropical cyclones. 1. **Consistency screening (matched grid specification)**: deterministic ACE2-ERA5 multi-member ensemble (forced over the same period as ERA5, ≥10 members recommended) vs. ERA5 on the identical grid and sampling interval. Criterion: ERA5 seasonal maximum median, 90th percentile, and 50-year return level fall within the emulator ensemble bootstrap 90% interval; relative bias within ±10% serves as an auxiliary reference, and bias lower or higher by >20% is recorded as systematic bias (direction not assumed a priori). Report both GEV and high-threshold POT. A 30-year run serves only as a trial run. Stochastic/generative comparisons use only ERA5-trained models (stochastic NeuralGCM, ArchesWeatherGen); ACE2S/HiRO-ACEv2 are trained on SHiELD/X-SHiELD and are not compared against ERA5. 2. **Station consistency check (not claimed as validating 3 s design gusts)**: ERA5 → station non-convective, non-tropical-cyclone gusts with held-out years for cross-validation; the 8-year Texas event list is insufficient, so longer ASOS series will be retrieved from IEM; explicitly state how quantile mapping handles values exceeding the observed maximum, and label the far tail as "unvalidated". 3. Proceed to Step 2 and trial design-value calculations only if Step 1 passes; if Step 1 fails and neither switching models nor downscaling can rescue it, shelve C and G. 4. Scale-up: once passed, run in batch across "station list × synthetic years" and deliver outputs to the EVA track. Numerical criterion thresholds are assumptions of this track (no literature basis).

Primary Criterion Determination (10-15, per parent Lead requirement; no existing result files modified)

1. **Pre-registered threshold: unspecified.** The revised protocol (Item 1 under "Verification Protocol" on this page) only stated that "the ERA5 seasonal maximum median, p90, and 50-year return values fall within the ensemble bootstrap 90% interval," without specifying how many grid points or metrics must fall inside to pass. No threshold is retrofitted post hoc.
2. **Why the interval is narrow.** The interval uses a two-level "member × forcing year" bootstrap estimating the uncertainty of the pooled 10-member × 14-season sample statistic, whereas ERA5 is a single 14-season realization. Because all 10 members share the same forcing and differ only in initial conditions, the pooled interval is naturally narrow and excludes the sampling variability of a single 14-season record itself. Under the current implementation, the primary criterion is therefore overly strict toward ERA5 and cannot be judged as failed solely on coverage counts. An apples-to-apples comparison with ERA5 is the distribution of statistics computed individually from each member's own 14-season record (the single-member range being computed by wx)—**this interval construction was modified after inspecting the results, constituting a post-hoc adjustment rather than a pre-registered rule**, and cannot serve as grounds for passing the pre-registered criterion.
3. **Determination: inconclusive.** Under the pre-registered criterion itself (unspecified threshold and interval construction mismatched to a single ERA5 realization), it is judged neither passed nor failed.

Grid-by-grid coverage (held-out 14 seasons, ensemble 90% interval):

Grid Point	Seasonal Max Median	Seasonal Max p90	GEV 50-yr	POT 50-yr
KMSP	Yes (−1.3%)	Yes (+3.5%)	Yes (−0.6%)	Yes (−0.8%)
KDSM	No (−6.2%)	No (−17.1%)	No (−28.9%)	No (−16.8%)
KOMA	No (−9.5%)	No (−16.7%)	No (−22.5%)	No (−15.6%)
KDTW	No (−3.4%)	Yes (−1.7%)	Yes (+4.6%)	Yes (+2.7%)
KPIT	No (+2.8%)	No (+6.0%)	Yes (+2.4%)	No (+6.0%)
KALB	No (+10.7%)	No (+9.7%)	Yes (+0.4%)	No (+12.1%)
DFW	Yes (+1.7%)	No (−5.8%)	No (−8.4%)	No (−4.5%)
AUS	No (−4.7%)	Yes (+0.1%)	Yes (−4.6%)	No (−4.9%)
Coverage	2/8	3/8	5/8	2/8

Auxiliary evidence (not pre-registered; formal report from wx): - Single-member range (held-out 14 seasons, ERA5 vs min–max across the 10 individual members, m/s): KDSM GEV50 19.0 / 12.6–14.1, POT50 16.1 / 12.7–13.8, p90 15.3 / 11.8–12.8; KOMA GEV50 17.5 / 12.1–15.1, POT50 16.0 / 11.8–14.3, p90 15.2 / 11.6–13.1—ERA5 exceeds all 10 members on all three metrics at both grid points (rank 11/11). GEV50 at the remaining six grid points falls within the member range; other out-of-range cases: KALB POT50 and p90 (ERA5 below all members), DFW p90 and AUS POT50 (above all members). Training-period GEV50 at KDSM and KOMA falls within the member range. - Strong-season synoptic type: The strongest held-out season at both KDSM and KOMA is 1998-11-10 (documented by NOAA/NWS as a severe winter storm with record-low pressure, verified by Lead against the NWS Sioux Falls page); the second strongest at KDSM is the 2009-12-09 blizzard (NWS Des Moines); 2008-10-26 is post-cold-front high wind; KOMA 2009-03-23 had a squall line that day but occurring after the peak hour, remaining undetermined; 2000-12-16 and 2010-01-25 lack collocated convective evidence. None of the top five strongest seasons show confirmable convection. - Interpretation: Each member's own 14-season estimate inherently carries the sampling variability of a 14-season record, making "ERA5 exceeding all 10 members" a better-matched comparison than the pooled ensemble interval; a single metric corresponds to a one-sided probability of ~1/11, though the two grid points are adjacent, share the same extreme event, and have highly correlated metrics that cannot be treated as independent trials. Under the auxiliary rule (still exceeding all 10 members after excluding convection → emulator tail suppression), the two plains grid points are judged as "emulator tail suppression, auxiliary-evidence level"; the pre-registered determination remains "inconclusive". 4. **Condition for advancing to the station stage**: No pass determination will be made on already-inspected data. Advancing requires writing the revised criterion (using the single-member distribution as reference and convective exclusion rules) and passing threshold (number of grid points and metrics) into a new pre-registration first, then testing on unseen data—such as the 2020/21–2021/22 held-out seasons (already covered by m01) or another set of unseen grid points.

Decisions Made

- 2026-10-17 **Remains Shelved**: post-hoc sensitivity exempting PK WND from Q3 yields an SS interval of [−0.09, 0.43] crossing zero; per the parent project rule that "passing depends on the Q3 specification," no further tests will be run →
- 2026-10-17 QC re-check mechanically "passed"; before reopening, added two post-hoc Q3 sensitivity checks (exempting PK WND from Q3, and dropping Q3 entirely), requiring the lower bound of SS to be >0 in both before switching back to Exploring and submitting for review first →
- 2026-10-17 Station-level track Shelved (set by root conversation); observational QC re-check pre-registration frozen at 10-17 18:49 and dispatched to wx, reopening only if v2 scores "pass" →
- 2026-10-16 Review verdict "upheld after revisions, report to Liang Hu for decision": added "Post-Review Corrections" per Mandatory Revisions 1–8 to the proposal and v2 decision (8-station spike contamination, KABR negative scale parameter, rephrasing SS as "no gain over baseline," GEV sensitivity 21/29 vs baseline 22/29 with SS −0.064, narrowed grid-cell conclusions, and three restart conditions); status remains Exploring →
- 2026-10-16 Station-stage v2 distributional transfer test **inconclusive** (hit rate 52%, SS −0.50 [−1.82, 0.03], R50 median deviation 0.22); point estimates indicate ERA5 carries essentially zero information for station distributions; no further protocol changes on the same dataset, escalated to parent project for decision →
- 2026-10-16 Parent project selected Option (b): record the v1 failure as-is and pivot to testing distribution-level spatial leave-one-out transfer; froze pre-registration v2 (22+22 candidate new stations, estimating `R` from held-out stations and benchmarking against an ERA5-free baseline) → ,
- 2026-10-16 Station-stage pre-registered test **failed** (`S > 0.10` at `0/5` stations, mean `S` CI `[−0.43, −0.02]`); `R` passed at `4/5` stations but is near unity by construction and does not count as positive evidence; the station interface halts at "grid-cell level passed, station level not established", awaiting parent-node decision on whether to switch validation tracks →
- 2026-10-16 Station-stage pre-registration frozen (criteria, stations, and time window locked prior to viewing any paired results) →
- 2026-10-15 Primary criterion determination: inconclusive (pre-registered threshold unspecified, interval construction mismatched); advancing to the station stage requires new pre-registration + unseen data → "Primary Criterion Determination" on this page,
- 2026-10-15 Interpretation of 10-member results and next steps →
- 2026-10-14 Literature review verdict: narrowed gap and revised protocol (unified event population, ensemble interval criteria, station stage downgraded to consistency check) → ,
- 2026-10-14 Project launch and validation protocol →

Attempts and Results

Date	What Was Done	One-Line Result
2026-10-14	Ran ACE2-ERA5 on Colab A100 for 30 years (1979–2008, single member, full year), annual max 10 m wind at 6 grid points vs. ERA5 (re-interpolated from WB2 1.5°)	Runnable: 2.6 min/simulated year, ~8.5 CU. Gumbel 50-year return level bias −14.1% to +17.8%, cross-grid median −2.1%, no grid point below −20%; mean wind median +2.4%; bias sign reverses east–west (KDTW biased low, KALB biased high). Counts only as a trial run: reference field resolution mismatched, mostly within training period, non-convective events not separated, no intervals
2026-10-14	Added DFW and AUS grid points to trial run (same 30-year single member)	Gumbel 50-year return level DFW +8.2%, AUS +14.0%; mean wind +5.8%, −1.8%; 3 of 8 grid points within ±10%, none below −20%. DFW grid point is ~69 km from the station
2026-10-16	Station-stage data reconnaissance (read-only, 0 CU)	All 15 airport ASOS stations corresponding to the grid cells have 1-minute peak wind archives on IEM dating back to at least 2000 (KAUS by 2005 at the latest); hourly METAR wind missing rate is <0.25%, though 1-week spot checks showed 1-minute missing rates of 37.6% at KDFW and 22.4% at KFSD; exact per-station sensor transition dates (Belfort 5 s / Vaisala 3 s / WMT702) were not yet retrieved; non-convective filtering can combine METAR weather codes + Storm Events + radar, none of which alone is a complete negative label
2026-10-16	Station-stage data audit (Colab)	Across 15 stations × 24 seasons, median 1-minute coverage is 92–96%; in 37 station-seasons the 1-minute peak exceeded METAR by ≥20 kn (28 isolated spikes); seasonal peak storm mechanisms: 152 non-convective / 68 convective / 134 ambiguous; HOMR cup-to-sonic transition occurred during 2005–2007

Date	What Was Done	One-Line Result
2026-10-16	Station-stage pre-registered test (Colab CPU, ~0.13 CU est.)	Failed: across the 5 primary stations, leave-one-out S spanned −0.22 to −0.06 with mean −0.11 [−0.43, −0.02]; R = 0.997–1.16; ERA5 vs station seasonal peak correlation was −0.25 to 0.33, sharing the same storm event in only 2–8/15 seasons; sensitivity station KICT was an exception at S = 0.59
2026-10-16	Station-stage v2 distributional leave-one-station-out transfer (Colab CPU)	Inconclusive: 29 stations (18 Plains, 11 non-Plains); abs(R10 − 1) ≤ 0.15 hit rate 52% (baseline 48%); SS −0.50 [−1.82, 0.03]; median abs(R50 − 1) 0.22; location mapping slope ≈ 0; worst station KABR R10 = 0.39
2026-10-17	Station-level observational QC re-check (pre-registered, Colab CPU ~0.11 CU est.)	Mechanically judged **passed**: across 29 stations, R10 hit rate is 28/29 (constant baseline 25/29), SS 0.392 [0.092, 0.584], R50 median deviation 0.062; post-QC cross-station correlation between ERA5 and station Gumbel location parameters is 0.75, scale 0.10; Q1 hit 13, Q2 hit 6, Q3 hit 162, Q4 hit 85 records, with 282/433 seasonal peaks unverified by ISD. The Lead found that applying Q3 literally to PK WND mistakenly deletes some genuine peaks; post-hoc sensitivity checks in progress
2026-10-17	QC post-hoc sensitivity: PK WND exempted from Q3	Changes only 8 station-seasons across 6 stations; SS 0.227 [−0.090, 0.434] → inconclusive; hit rate 28/29 vs. baseline 27/29; Plains SS 0.07
2026-10-17	QC post-hoc sensitivity: dropping Q3 entirely	Identical station-season by station-season to exempting PK WND from Q3 (SS 0.227 [−0.090, 0.434], inconclusive); the entire effect of Q3 comes from those 8 PK WND station-seasons

Date	What Was Done	One-Line Result
2026-10-16	Unseen-grid-cell pre-registered test (frozen criteria, 7 valid cells after excluding lake cell KMKE): 10-member rerun across 14 held-out seasons	**Primary criterion passed**: on GEV50, ERA5 fell outside the 10-member range at 2/7 cells (KFSD above all members, KSTL below all members), literally satisfying the ≤2 threshold. Auxiliary: POT50 and p90 each had 3/7 outside range; cross-cell median of member-median GEV50 bias was +4.7%. Plains KFSD again had ERA5 above all members (GEV50 bias −23%, training period −17%), matching the sign of KDSM/KOMA. On Plains surface pressure anomalies, ACE2's deepest 10% seasons were deeper than ERA5 (−39.8 vs −37.7 hPa), showing no cyclone weakening. All sessions stopped; member session ran 15.9 h (~87 CU est.)
2026-10-15	Formal 10-member round (ACE2-ERA5, Jan–Oct 1979 initial conditions, m02–m10 through 2010-04), 14 held-out seasons and 15 training seasons, 8 grid points cold season, ensemble bootstrap intervals (computed on Colab)	Held-out GEV50 bias median −2.6%, 6/8 grid points within ±10%; KDSM −28.9% and KOMA −22.5% biased low; training-period KALB +22.6% biased high. Coverage within ensemble 90% interval: GEV 5/8, seasonal max median 2/8, POT 2/8 (interval excludes ERA5 sampling error and is overly strict). ~94 CU, all sessions stopped
2026-10-14	Formal round first member m01 (1979-01 initial condition, 1979–2022), 0.25° ERA5 conservatively regridded reference, 8 grid points cold season; completed and stopped at 21:34	Predictions confirmed to include 10 m wind (19:42 warning harmless). Across 16 held-out seasons: cross-grid seasonal max median bias +0.0%, p90 −0.6%, GEV 50-year return level median −8.6% (KOMA −25% to KALB +26%), POT 50-year return level median −5.1%; across 23 training seasons, GEV median −2.6%. Single member, no ensemble interval; official initial conditions only available for 1940/1950/1979/2001/2020, none for 1995. ~13.6 CU

Date	What Was Done	One-Line Result
2026-10-14	Formal round preparation: ERA5 0.25° conservatively regridded directly to ACE2 1° grid (ACE2 1° training bucket requires a paid GCP project, not used), extraction completed for 42 cold seasons × 8 grid points; 16 held-out cold seasons (1996/97–2009/10 and 2020/21–2021/22)	Relative to the old 1.5° interpolated reference, the new reference shifts 8-grid mean wind by −6.7% to +6.1% and cold-season sample maxima by −3.7% to +5.7%: the reference specification alone has a several-percentage-point effect. Scaled back at 19:45 under the ≤20 CU cap: 10-member controller stopped, keeping only the 1979–2022 first member monitored locally by budget_watch.py, downloading point series and stopping upon completion or force-stopping by 22:30 at latest; yielded only a single-member preview in that round, later superseded by the parent project's quota correction to complete all 10 members (see Open Items and Next Steps)
2026-10-14	Recomputed existing single-member 30-year trajectory under revised protocol: cold-season (Oct–Mar, 29 seasons) maxima, seasonal GEV, seasonal POT (p95 threshold, 72 h declustering, assumed parameters), annual GEV, local CPU, 0 CU	Across 8 grid points, seasonal GEV 50-year bias median −3.8% (−10.3% to +12.7%), seasonal POT median −3.8% (−10.1% to +12.4%), no grid point exceeding ±20%, with both positive and negative signs. Annual GEV reaches +37%/+40% at DFW/AUS (whereas Gumbel on the same data is only +8%/+14%): with a 30-year sample, the fitting method alone can flip the verdict, supporting the switch to seasonal/POT and ensemble intervals. Cold season ≠ non-convective separated. Ensemble intervals not yet evaluated
2026-10-14	Checked Paciorek 10,000-year ensemble; ran ACE2S-SHiELD+ (stochastic) and ACE2-SHiELD (deterministic) for 10 years each (1979–1988), 8 grid points against ERA5; verified ERA5 10 m wind definition	10,000-year ensemble does not publish wind outputs (configuration saves only TMP2m and PRATEsfc). Stochastic version runs and outputs 10 m wind, but its training target is SHiELD and its training recipe differs from the deterministic version, so it is not a pure stochastic toggle; stochastic vs. deterministic 50-year values both increase and decrease across grid points (−20% to +5%), so stochasticity cannot be said to fix the wind tail; neither model can be evaluated under ERA5 criteria. ERA5 land 10 m wind: when z0>0.03 m, extrapolated from 40 m using 0.03 m open-terrain roughness (IFS Cy41r2 Part IV §3.10.2, PDF pp. 52–53). ~7.6 CU

Date	What Was Done	One-Line Result
2026-10-14	Grok quick check + verification	ACE2-ERA5 is top candidate (native 10 m wind output); no direct precedent for estimating design wind speeds with AI emulators; claim of low bias in extratropical cyclone winds pending verification

Parameter Sources

Parameter	Value	Source or "Assumption"	Places Checked
Bias criterion ±10% / systematic low bias >20%	—	Assumption (experimental criterion for this track, not a literature value)	—
ERA5 land 10 m wind diagnostic	Extrapolated from 40 m using 0.03 m open-terrain roughness when model z0>0.03 m, otherwise uses model roughness	IFS Cy41r2 Part IV §3.10.2 (PDF pp. 52–53)	ECMWF documentation; User Guide §9.3
Compute and storage locations	Workstation does not run computations or store compute data: model runs, ERA5 extraction, and GEV/POT/bootstrap statistics all run on Colab (or CRC/TACC); workstation holds only orchestration scripts and small results <200 MB per item, figures, and reports; temporary storage >1 GB requires reporting to the parent project first; legacy compute data stored on TACC	Lab-wide new rule by Liang Hu on 10-14, relayed by parent project on 10-15	—
Compute boundary	No single-experiment cap on Colab; report to parent project first if expected >100 CU; stop when finished	Liang Hu's exact words 'use Colab heavily at this stage, unlimited CU', corrected by parent project on 10-14 (superseding ≤20 CU)	—

Open Items and Next Steps

- 10-16 Station data audit completed (15 stations × 24 seasons; primary stations KDSM/KOMA/KFAR, controls KIND/KBDL, sensitivity KFSD/KICT, dropped KAUS/KPIT) →.
- 10-16 Station-stage pre-registration v1 frozen → : primary window 2007/08–2021/22 (all-ultrasonic period across the 5 stations); METAR non-convective seasonal peak gusts; Track A leave-one-season-out linear mapping skill S (baseline: training mean), Track B odd/even split-half mean–SD mapping Gumbel 10-year ratio R; pass criteria = S > 0.10 at ≥4/5 stations AND mean S CI lower bound > 0 AND |R − 1| ≤ 0.15 at ≥3/5 stations. Dispatched to research assistant (queued, Colab <10 CU).
- 10-16 Station-stage v1 failed and v2 was inconclusive. Awaiting parent project decision: (a) wrap up at the grid-cell level (abandoning the station track, requiring 6-reviewer panel + Liang Hu); (b) pivot to a new "exposure correction" problem (changing the scientific question, requiring Liang Hu). G remains deferred.
- 10-16 Unseen-grid-cell validation primary criterion **passed** (2/7 outside range). Caveats: Plains KFSD shows the same-sign low bias (22–29% low) as KDSM/KOMA, pointing to potential regional tail suppression in the Plains (cyclone intensity is not weaker, so the issue likely lies in the pressure-to-surface-wind mapping); rank counts are not a formal significance test; 50-year return levels are extrapolated from 14 seasons. Next step per protocol is the station stage: first dispatch wx for a read-only station data survey (IEM ASOS long records, sensor transitions, convective labeling rules), then write the station-stage pre-registration with Plains stations examined separately.
- 10-16 00:xx Regional field uploads canceled starting from m03 to control CU usage (m01 measured 1.72 h / ~9.4 CU, projecting 98.7 CU total near the 100 CU reporting threshold; immediate supervisor did not approve 115 CU, as >100 CU authorization rests with the parent project); auxiliary pressure statistics (Amendment 1 surface pressure anomaly) switched to in-Colab computation returning compact tables. If re-estimation still exceeds 100 CU, pause and report upward before crossing the threshold rather than unilaterally dropping members.
- 10-15 19:xx Dispatched wx: per the pre-registration (frozen prior to launch), rerun 10 members across 8 unseen grid points (KFSD, KFAR, KICT, KSTL, KMKE, KIND, KCLE, KBDL) and archive CONUS-domain 10 m wind and sea-level pressure fields (stored on TACC), with the rank count of ERA5 among the 11 values on held-out GEV50 as the primary criterion; ~75–90 CU, pause and report if >100 CU. At 19:02 wx pre-check found no PRMSL in the checkpoint; Pre-registration Amendment 1 was added before launch (switching the auxiliary cyclone indicator to surface pressure anomaly while keeping the primary criterion unchanged) and re-dispatched at 19:10. Launched at 19:17: the nearest grid cell to KMKE is predominantly lake surface (land fraction 0.20) and was excluded without replacement per the frozen rule, leaving 7 valid grid points; passing count thresholds remain unchanged (≤2 out-of-range grid points to pass, ≥3 grid points above all members for tail suppression); ERA5 reference extracted on Colab, orchestrator orchestrate_new.py running 10 members sequentially in background, expected to finish around noon on 10-16.
- 10-15 18:50 Dispatched wx: single-member rank (whether ERA5 falls outside the 10 individual member estimates) + KDSM/KOMA held-out strong-season event identification (whether convective events are mixed in), <5 CU. See criteria.

- Formal round (ideas; Colab policy corrected to no cap, report first if >100 CU): after the first member (1979–2022) completed, add 9 members (initialized Feb–Oct 1979) integrated through 2010-04, giving 10 members covering 14 held-out cold seasons (1996/97–2009/10), estimated at ~80 CU total. Starting 10-14 21:40, controller_2010.py ran the 9 members sequentially; at 22:13, a single colab polling call timeout during member 2 was misclassified as fatal by the controller, stopping the session (~3 CU wasted) and entering HALTED with 0 new members completed. Re-dispatched to ideas at 10-15 02:35: change polling timeouts to retry, resume m02–m10 (without re-running m01), and compute statistics inside Colab; estimated total <100 CU. Resumed at 10-15 02:35, but at 02:52 the controller again erroneously stopped the session during m02 preparation due to an exec timeout (~4.5 CU wasted across the two stops, 0 new members); at 03:40 instructed ideas to move preparation and execution to background processes on Colab with marker files and retry-only on polling timeouts (using colab-keepalive), running m03–m10 sequentially once m02 succeeds. At 03:46 m02 launched in the background via colab-keepalive and reached window 38/911 by 03:51; at ~80 min/member, the 9 members were projected to finish around 10-15 16:00. At 05:20 m02 finished and shut down, and m03 began preparation; including preparation (~95 min/member), the remaining 8 members are projected to finish around 10-15 18:00, with total usage projected at ~86.6 CU. At 06:53 m03 completed and was preserved (the controller vanished mid-run, so ideas manually wrapped up and stopped the session; cause under investigation), and m04 was started by the new controller. As of 10-15, this node belongs to the AI Weather project, with new tasks dispatched to wx; the running ideas 10-member task will run to completion as planned. Lead fallback follow-up at 10-15 10:44 retained. Original plan: use ACE2's own 1° ERA5 training data as reference, focus on held-out periods (1996–2010, 2020–2022), ≥10-member ensemble intervals, cold-season non-tropical-cyclone grid points, GEV+POT. Once results are in, issue a judgment on whether it is worth continuing and report upward.
- Hierarchy rules (root Lead 10-14, 5-level cap): this node has been migrated under "Climate and Design Wind Speeds" (level 2). No level-3 sub-projects will be opened under this node; subdivisions are handled as tasks within this conversation or reported to the parent "Climate and Design Wind Speeds" node Lead to open a peer node (which will also open G when the time comes); downward only two hops remain (this Lead → bot → tool), and task briefs must not allow tool bots (gemini-worker, computeops, litwiki, etc.) to delegate further. Running delegations to ideas will report back here as usual.
- Which emulator is usable and whether it outputs 10 m wind → to be decided after ideas reports back
- Station data (EVA Lead reply on 10-14): the EVA track does not run AI emulators or ERA5 → station conversions, so there is no duplication. Ready-made read-only data consists solely of the four-station Texas Cook ASOS gust event list (≥20 kt, Class=5 treated as thunderstorm), where DFW is most suitable, followed by AUS, with HOU excluded. Limitations: it is an event list rather than annual maxima; gust averaging duration varies by sensor (Belfort 5 s / Vaisala 3 s) with no replacement history; QC can erroneously delete peaks (H7ZG33PU p.9); covers only 8 years. → For the station conversion step: use EVA data for DFW/AUS, and have ideas retrieve public ASOS data from IEM for central/northeastern stations; attach conversion errors when delivering results back to EVA, and self-check using their v2.7 checker.
- Literature step 2 Deep Research → write the second task brief after Grok quick check returns

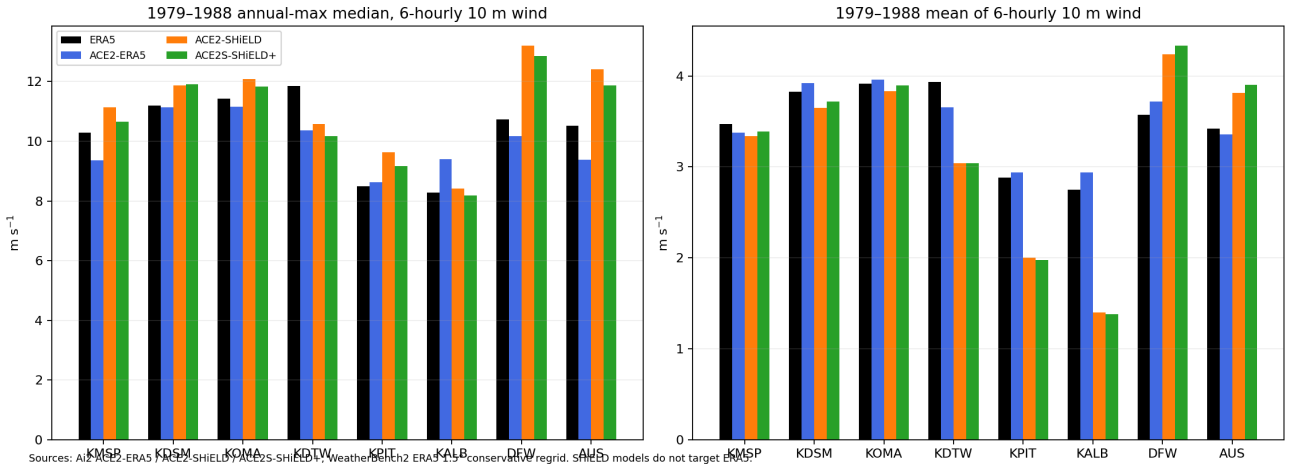
Literature Basis

- Paciorek & Cooley 2026 BAMS 107(5), DOI 10.1175/BAMS-D-25-0178.1 (abstract): ACE2 10,000-year ensemble for extreme precipitation/temperature; authors call for testing whether emulators are fit for purpose
- T.-C. Chen, Collet, Di Luca 2024 IJOC 44(3):729–747, DOI 10.1002/joc.8339 (abstract): ERA5 North American extratropical cyclone 10 m wind top 5% NMB −10.2%
- Dacre et al. 2026 BAMS 107(1), DOI 10.1175/BAMS-D-25-0129.1 (abstract): 7 ML weather models underestimate cyclone minimum pressure by >5 hPa (at 10 days) and produce weaker peak 10 m winds, whereas IFS has no bias
- ACE2: Watt-Meyer et al. arXiv 2411.11268 (HTML full text Methods, §2.2.3, §2.2.6): 10 m horizontal wind is a prognostic variable; TC maximum 10 m wind is accurately simulated relative to ERA5; individual responses to SST/CO2 are not fully realistic (abstract)
- HiRO-ACEv2: Kwa et al. arXiv 2610.17693 abstract: near-surface wind distribution matches up to the 99.9999th percentile in the X-SHiELD model world (lead)
- : ArchesWeather/Gen 45-year AMIP-style run; deterministic models regress toward the mean while generative models may preserve tails (preprint + WeChat post, full text unread)
- : NeuralGCM (Nature 2024, DOI 10.1038/s41586-024-07744-y) abstract states it cannot extrapolate to substantially different future climates
- EVA cards: mixed-climate EVA, thunderstorm/non-thunderstorm separation, 18 m/s threshold (PDF pp. 2–3, 5); `KGS2TG8X` (climate change and extreme storms, designload_step5)

Liang Hu Notes

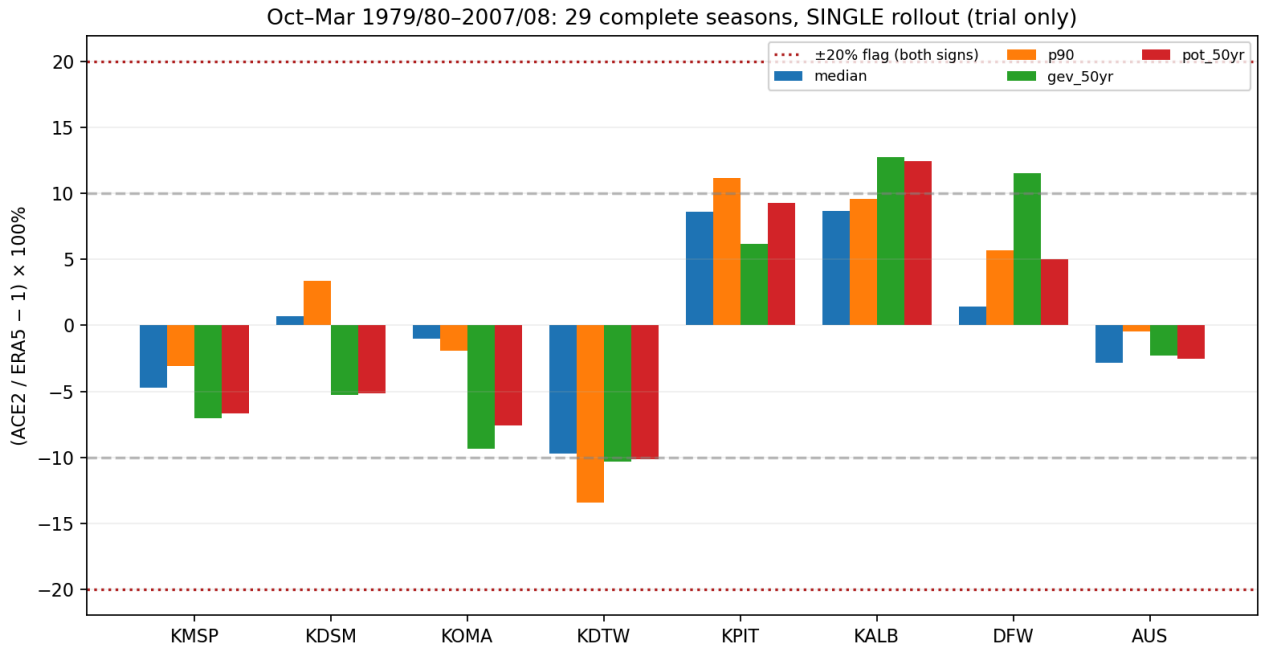
- 10-14: Approved C; validate extremes first on Colab; Colab balance is ample at this stage, so CU is unlimited (supercomputers still follow existing rules). [Lead note (10-14 correction): per Liang Hu's exact words, Colab has no single-experiment cap; if a single experiment is expected to exceed 100 CU, pause and report to the parent project first; must run colab stop when finished; dispatch only one executor per round. The previous '≤20 CU' limit is voided]
- 10-14: G serves as Stage 2 of C; "This research can be done, though its scientific value is moderate—it is fine for publishing a paper. Currently there is no evidence proving that climate change will lead to an increase in design wind speeds."
- 10-14 Overall guideline: the ultimate goal is at-scale forecasting and hindcasting.

Key figures



Sources: Ai2 ACE2-ERA5 / ACE2-SHiELD / ACE2S-SHiELD+; WeatherBench2 ERA5 1.5° conservative regrid. SHiELD models do not target ERA5.

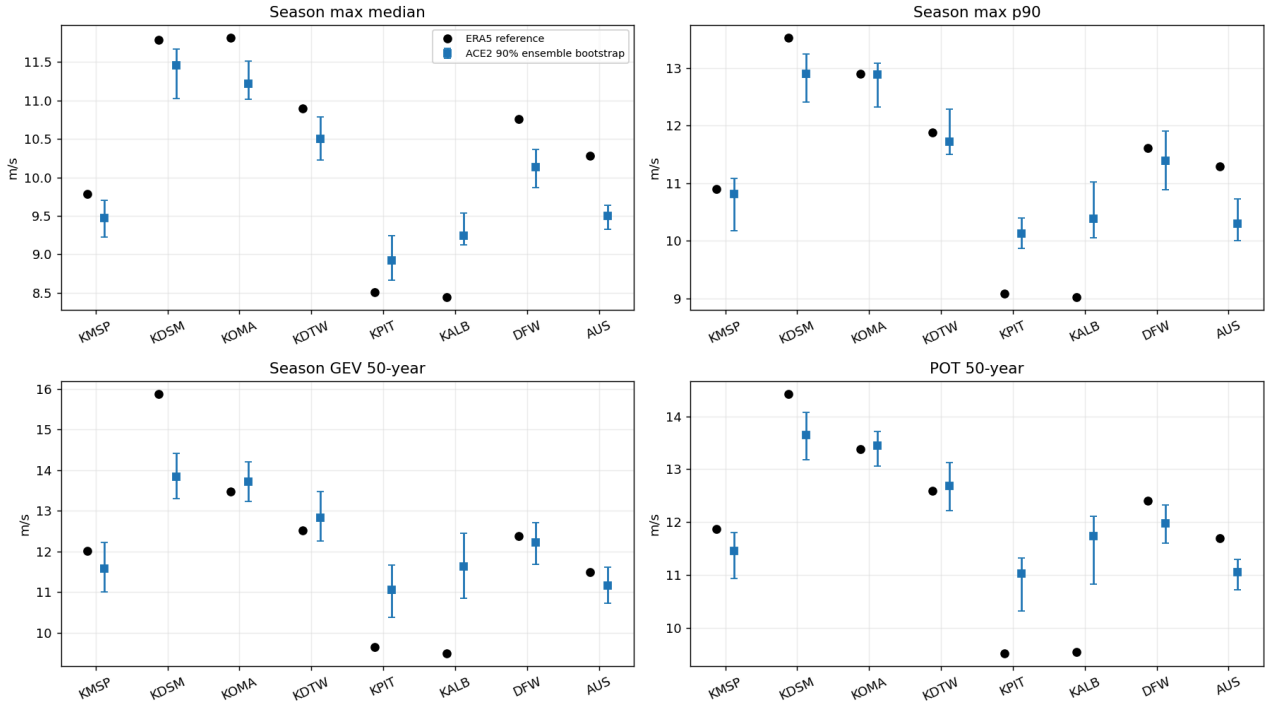
paired shield decade



Source: ACE2-ERA5 (Ai2) 10m U/V; ERA5 via WeatherBench2 1.5° conservative → 1° linear. 6-hourly; no event-type filtering.

cold season trial

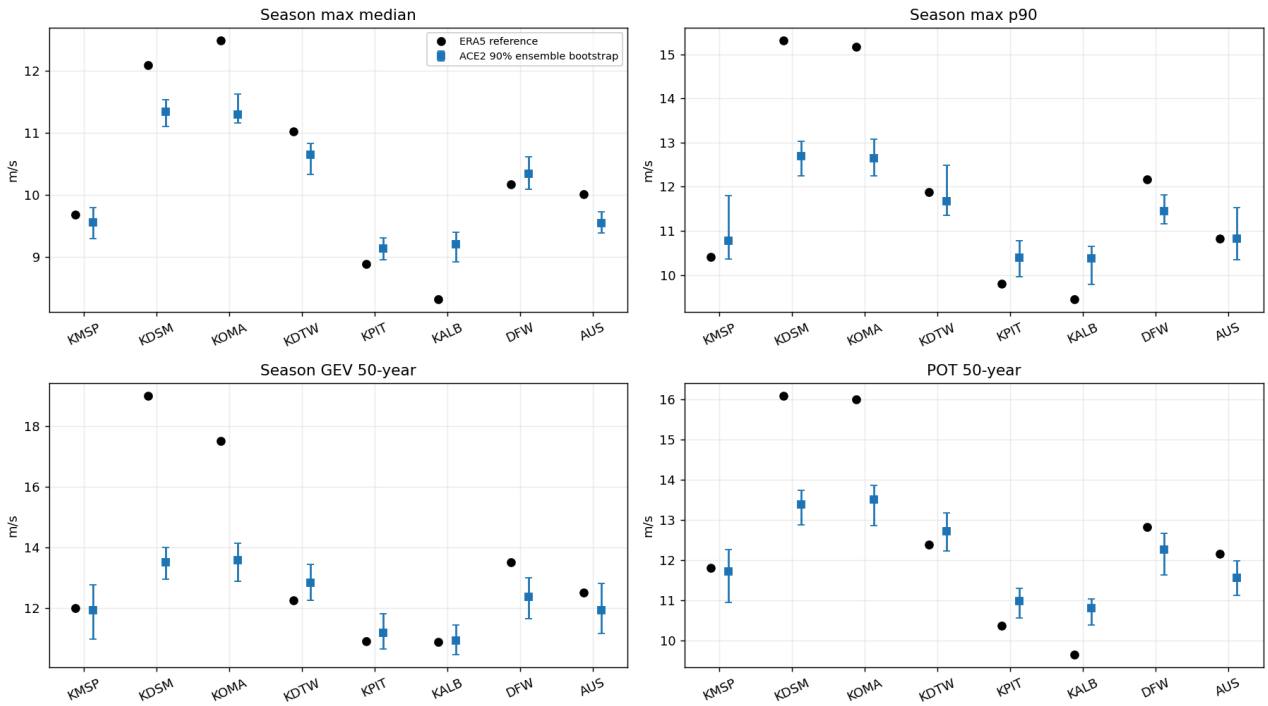
ACE2–ERA5 training: 15 complete Oct–Mar seasons; 10 monthly ICs



Source: Ai2 ACE2-ERA5 HF; ECMWF ERA5 via WeatherBench2 0.25° conservatively remapped. Cold-season proxy, not event-classified.

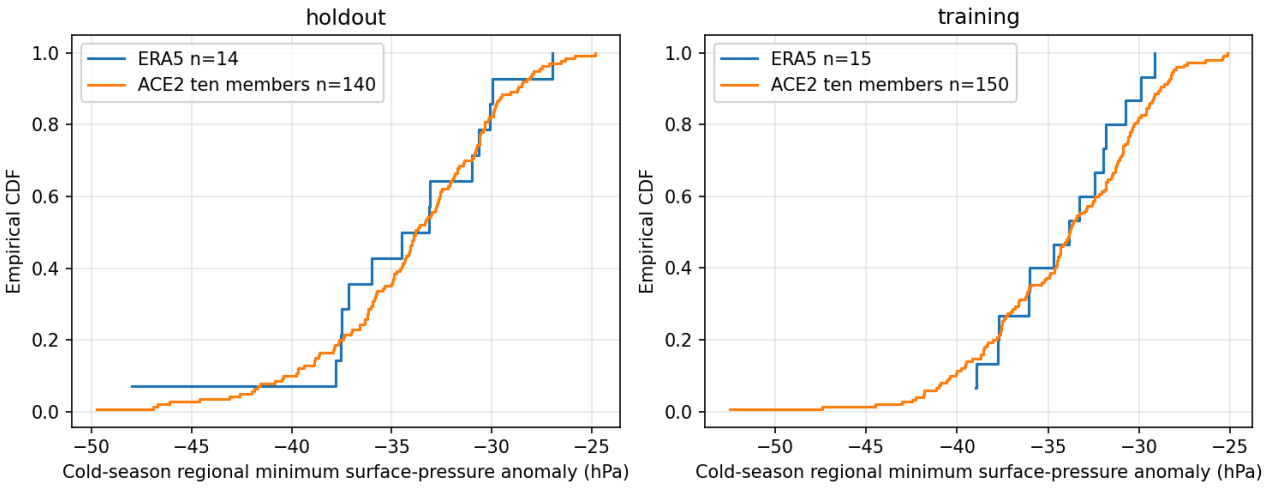
training comparison

ACE2–ERA5 holdout: 14 complete Oct–Mar seasons; 10 monthly ICs



Source: Ai2 ACE2-ERA5 HF; ECMWF ERA5 via WeatherBench2 0.25° conservatively remapped. Cold-season proxy, not event-classified.

holdout comparison



pressure anomaly cdf